



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.DS.6C

Mean Absolute Deviation

Lesson 8-3 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can find the mean absolute deviation (MAD) to describe how spread out data is.

Language Objective: I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean Absolute Deviation

Deviation

Absolute Value

Spread

Data distribution

Variability



Tap any word to see what it means and a picture.

1

The average distance of each value from the mean is the

.

2

How far a single value is from the mean is its

.

3

The distance of a number from zero, always positive, is its

.

4

How far apart the data values are from each other is the

.

5

The way data values are spread out is the

.

6

How much the data values differ is the

.

Reflect — Exit Ticket

Data set: 4, 8, 6, 10, 2. The mean is 6. What is the MAD?

- A. 2.4
- B. 6
- C. 0
- D. 12

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Mean Absolute Deviation
2. **Notes 2:** Deviation
3. **Notes 3:** Absolute Value
4. **Notes 4:** Spread
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Mean Absolute Deviation is skipping the key idea: "MAD = the average of how far each number is from the mean (always a positive distance)." — always check your work against this rule before you submit.*
8. **Try It 1:** B. 2 — *Add the distances: $3 + 1 + 1 + 3 = 8$. Divide by the 4 games: $8 \div 4 = 2$. The MAD is 2 points.*
9. **Try It 2:** A. Shooter A, because a smaller MAD means scores stay closer to the average — *A smaller MAD means the scores hug the mean. Shooter A's MAD of 2 is smaller than Shooter B's 6, so Shooter A is more consistent.*
10. **Exit Ticket:** A. 2.4 — *Deviations: $-2, 2, 0, 4, -4$. Absolute deviations: $2, 2, 0, 4, 4$. Sum = 12. MAD = $12 \div 5 = 2.4$.*